

Dataset Description

AI-Assisted Data Analytics Final Project

Dataset Title	Customer Personality Analysis / Marketing Campaign Dataset
Dataset Source	Kaggle — kaggle.com/datasets/rodsaldanha/arketing-campaign (by Rodrigo Saldanha)
Business Domain	Marketing / Customer Analytics
Rows (Records)	2,240 (raw) → 2,236 after cleaning
Columns (Variables)	29 (raw) → 27 after cleaning
Target Variable	Response (binary: 1 = accepted the last marketing campaign, 0 = did not)

Variable Descriptions

The table below describes every variable retained in the cleaned dataset. Two constant columns present in the raw file (Z_CostContact, Z_Revenue) carried a single repeated value across all records and were removed during data cleaning as they add no analytical value.

#	Variable	Type	Description
1	ID	Numerical (ID)	Unique identifier assigned to each customer.
2	Year_Birth	Numerical	Customer's birth year; used to derive age.
3	Education	Categorical	Customer's education level: Basic, 2n Cycle, Graduation, Master, or PhD.
4	Marital_Status	Categorical	Customer's marital status: Single, Married, Together, Divorced, or Widow.
5	Income	Numerical	Customer's yearly household income in US dollars.
6	Kidhome	Numerical	Number of small children in the customer's household.
7	Teenhome	Numerical	Number of teenagers in the customer's household.
8	Dt_Customer	Date	Date the customer enrolled with the company.
9	Recency	Numerical	Number of days since the customer's last purchase.
10	MntWines	Numerical	Amount spent on wine products in the last 2 years.
11	MntFruits	Numerical	Amount spent on fruit products in the last 2 years.
12	MntMeatProducts	Numerical	Amount spent on meat products in the last 2 years.
13	MntFishProducts	Numerical	Amount spent on fish products in the last 2 years.
14	MntSweetProducts	Numerical	Amount spent on sweet products in the last 2 years.
15	MntGoldProds	Numerical	Amount spent on gold/premium products in the last 2 years.

#	Variable	Type	Description
16	NumDealsPurchases	Numerical	Number of purchases made using a discount/deal.
17	NumWebPurchases	Numerical	Number of purchases made through the company's website.
18	NumCatalogPurchases	Numerical	Number of purchases made using a catalog.
19	NumStorePurchases	Numerical	Number of purchases made directly in physical stores.
20	NumWebVisitsMonth	Numerical	Number of visits to the company's website in the last month.
21	AcceptedCmp1-5	Categorical (binary)	Five separate flags (1/0) indicating whether the customer accepted the offer in marketing campaigns 1 through 5.
22	Complain	Categorical (binary)	Whether the customer complained in the last 2 years (1 = yes, 0 = no).
23	Response	Categorical (binary)	Target variable: whether the customer accepted the offer in the most recent campaign (1 = yes, 0 = no).

Why This Dataset Was Selected

This dataset was chosen because it realistically represents a customer relationship management (CRM) database that a marketing team would maintain. It combines demographic attributes (age, education, income, family structure), behavioral data (spending across six product categories, purchase channels), and historical campaign responses in a single table. This combination makes it possible to study not just who the customers are, but how they behave and how they have responded to marketing in the past — which is exactly the kind of data needed to predict future campaign response.

Business Problem / Analytical Objective

Marketing teams have limited budgets and cannot afford to target every customer in every campaign. The core business problem is: **which customers are most likely to respond positively to a new marketing campaign, and what customer characteristics drive that response?** By predicting the probability that a given customer will accept an offer (the Response variable), the marketing team can prioritize high-probability customers, improve campaign ROI, reduce wasted contact costs, and design more personalized offers based on the customer segments and spending patterns uncovered during exploratory analysis.

Why the Dataset Satisfies the Project Requirements

- Structured, tabular data with 2,240 original records and 29 original variables, comfortably exceeding the minimum of 1,000 rows and 8 variables.
- Belongs to a recognized real-world business domain (Marketing / Customer Analytics).
- Contains a mix of numerical variables (Income, Recency, spending amounts) and categorical variables (Education, Marital_Status), with meaningful relationships between them (e.g., income vs. spending, family size vs. purchase channel).
- Includes a clearly defined, appropriate target variable (Response) suitable for a binary classification problem.

- Contained genuine real-world data quality issues that required cleaning: 24 missing Income values, implausible birth years (e.g., 1893, 1899, 1900), an extreme income outlier (666,666), inconsistent Marital_Status categories ("Alone", "Absurd", "YOLO"), an incorrectly typed date column, and two constant, non-informative columns.
- Supports rich feature engineering (e.g., customer age, total spending, total children, tenure with the company) and predictive modeling.
- Naturally supports an interactive dashboard with KPIs such as campaign acceptance rate, average customer spending, and channel usage, all of which can be filtered by demographic slicers.